Self-Ensemble Learning Based on Energy Attention and Dual-Branch Guidance for Hyperspectral Image Classification

Wenjie Li, Ronghua Shang, *Senior Member, IEEE*, Jingyi Ding, *Member, IEEE*, and Jie Feng, *Senior Member, IEEE*

***Abstract*—Due to the complex spatial heterogeneity and spectral heterogeneity of hyperspectral images (HSIs), applying the self-ensembling learning with a single noise perturbation in HSI classification (HSIC) is challenging. Therefore, this article proposes a self-ensemble learning based on energy attention and dual-branch guidance (EDSEL) for HSIC. First, this article designs a neural network based on energy attention to extract spatial and spectral features of HSIs. The spectral and spatial attention is optimized through the energy function between pixels in the spectral and spatial dimensions to preserve multiscale spatial structure information and obtain discriminative features of ground objects with complex spatial distributions. Second, EDSEL uses a teacher-assisted loss (TAL) function for the training process of labeled samples. The classification loss function incorporates the alignment information between the teacher and student networks. This process assists the student network in learning from the teacher network's knowledge and in extracting features that distinguish the labeled samples. Finally, EDSEL designs a dual-branch guidance strategy for consistency learning of unlabeled samples. Two branches use static and temporal noise to help the network obtain consistent representations under different noise conditions, learning the essential spatial–spectral features, and enhancing the classification performance. The experimental results on three HSI datasets show that the EDSEL performs better than the seven advanced algorithms.**

***Index Terms*—Consistency learning, deep learning, hyperspectral image (HSI) classification, self-ensemble learning, spectral–spatial attention.**

I. Introduction

With the continuous development of imaging spectroscopy technology, hyperspectral image (HSI) based on wide spectral range acquisition has gradually received widespread attention [1]. This type of image consists of multiple spectral bands beyond the visible light range, containing richer and more detailed descriptions of object information [2]. Therefore, HSIs have been widely used in military reconnaissance [3], national defense security [4], agricultural monitoring [5], and mineral investigation [6]. The HSI classification (HSIC) task is to accurately classify the pixel vectors of HSIs based on their rich spatial and spectral information. However, the abundant bands lead to data redundancy and increased processing complexity [7], and labeling HSIs is a time-consuming and laborious task, which has become the main challenge for HSIC [8].

Traditional machine learning methods utilize domain experts and professional skills to design basic features manually, providing key information for image classification. The commonly used machine learning methods include random forest (RF) [9], K-nearest neighbor (KNN) algorithm [10], and scale invariant feature transform [11]. Xia et al. [12] constructed an ensemble of RF classifiers based on enhanced rotation to generate classifiers that balance accuracy and diversity. Tu et al. [13] used the KNN algorithm to distinguish between different superpixels and different categories of pixels within superpixels, improving the results of HSIC. Although the above methods have been widely applied in remote sensing images [14], [15], the core bottlenecks conflict with the inherent characteristics of hyperspectral data to a certain extent. HSIs contain hundreds of contiguous spectral bands with strong nonlinear correlations among bands, as well as severe phenomena of within-class spectral variability and between-class spectral similarity. Nevertheless, manually designed features can only capture predefined shallow spectral patterns and fail to adaptively mine fine-grained discriminative information in the high-dimensional spectral space. Meanwhile, the design of such features heavily relies on professional prior knowledge from remote sensing spectroscopy and geoscience, leading to poor generalization ability. Once the sensor type, observation area, or land cover category changes, the classification performance of handcrafted features will drop drastically, making them unable to adapt to the complex and variable application scenarios of hyperspectral data.

Compared with machine learning methods, deep learning methods can map and characterize data features without prior knowledge of the input data. Therefore, deep learning methods have attracted researchers' attention. Deep learning methods use stacked neural networks to extract spectral and spatial features of HSIs in spectral and spatial dimensions to classify different pixels. Common neural network structures include the convolutional neural network (CNN) [16], the generative

Received 14 December 2025; revised 16 March 2026; accepted 17 April 2026. Date of publication 22 April 2026; date of current version 8 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62576260, Grant 62176200, Grant 12326617, Grant 62271374, and Grant U2570205. *(Corresponding author: Ronghua Shang.)*

The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an, Shaanxi 710071, China (e-mail: wjli_3@stu.xidian.edu.cn; rhshang@mail.xidian.edu.cn; jyding@xidian.edu.cn; jiefeng0109@163.com).

Digital Object Identifier 10.1109/TGRS.2026.3686411

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adversarial network (GAN) [17], and the Vision Transformer [18]. CNN obtains local features of images through receptive fields and semantic features through hierarchical methods. Li et al. [2] utilized a nonlocal fusion method to integrate features from different CNN layers, achieving more comprehensive results. Sun et al. [19] proposed an end-to-end fully convolutional segmentation network to improve the generalization performance of the network. However, the inherent limitations of CNN are in significant contradiction with the characteristics of hyperspectral data, which limits the upper bound of classification performance. The convolution operation of CNN is strictly local and can only capture neighborhood features within the receptive field, failing to model the long-range spatial dependencies and global spectral consistency of large-scale homogeneous ground objects in HSIs. This easily leads to fragmentation and speckle-like noise in the classification results of the same category. In addition, the number of parameters of the CNN increases linearly with the number of spectral bands. Faced with hundreds of input bands in hyperspectral data, many high-quality labeled samples are required to fit the nonlinear spectral–spatial mapping relationship. However, the annotation of HSIs requires dual professional knowledge of ground object spectroscopy and geospatial information, resulting in extremely high labeling costs and scarce available samples. Under small-sample scenarios, CNN is prone to severe overfitting, and the classification accuracy for rare ground objects with extremely low sample proportions drops sharply.

GAN reduces the need for labels to a certain extent through the game between the discriminator and the generator. Zhong et al. [20] constructed a semisupervised GAN to alleviate the shortage of labels while integrating spectral and spatial attributes into the network to learn discriminative features. Jiang et al. [21] designed SCGAN to enhance the distinction between anomalies and backgrounds through background homogenization and anomaly saliency, reducing the need for anomaly labels. However, GAN also has certain limitations. Most generators and discriminators are built based on convolutional structures, which still cannot effectively capture the long-range spatial–spectral dependencies of hyperspectral data. Meanwhile, GAN cannot explicitly model the strong correlations among hyperspectral bands, resulting in poor spectral continuity of generated samples and extremely weak generalization performance in scenarios such as within-class spectral variability. Therefore, researchers turned to the Transformer for long-range information. Li et al. [22] proposed a biplane branch Transformer that combines local and global features to form multidimensional and multiscale features. Yang et al. [23] proposed a quaternion transformation network to extract local semantic features and global representations from the perspective of the quaternion domain. However, the above methods still require a large amount of labeled data to assist the classifier learning.

The labeled data used in the aforementioned methods typically require significant effort and expense. In contrast, unlabeled data are easier to acquire and often available in large quantities. As a result, researchers have increasingly focused on leveraging vast amounts of unlabeled data to assist in training classifiers. This convergence has led to semisupervised learning (SSL) [24]. Depending on how unlabeled data are utilized, SSL methods can be categorized into graph-based, pseudolabeling, and consistency regularization methods. Graph-based methods primarily construct similarity graphs and perform label inference by integrating topological structure and feature learning, thereby enabling the classification of unlabeled data. For example, Yang et al. [25] dynamically constructed multiscale feature graphs to exploit the rich information while updating pixel-level and region-level representations to capture semantic information. Xiao et al. [26] designed a dynamic graph evolution mechanism and adopted message-passing schemes to capture both local attention in the graph and long-range dependencies. Kotzagiannidis and Schönlieb [27] introduced multiview learning into the HSI domain and utilized label information to enhance the contribution of superpixel features in the embedded graph, reducing the need for extensive parameter tuning. However, graph-based methods generally suffer from high computational complexity due to the intensive matrix operations involved.

Pseudolabel learning methods first use labeled samples to train a classifier. Then, the trained classifier assigns high-confidence pseudolabels to unlabeled data. These pseudolabeled samples are subsequently used together with the labeled samples for further training [28], [29]. Pseudolabel learning can effectively leverage large amounts of unlabeled data to participate in training, thereby improving model performance. Wu and Prasad [30] proposed a constrained Dirichlet process mixture model, which generates higher-quality pseudolabels by incorporating spatial information. Wei et al. [31] combined loss reweighting and dynamic feature clustering algorithms to generate pseudolabels, facilitating the training of unlabeled samples. Seydgar et al. [32] introduced an uncertainty-aware probabilistic pseudolabel generation framework, assuming that the uncertainty of labeled samples follows a Gaussian distribution and using this distribution to generate high-confidence pseudolabels for unlabeled samples in the target domain. In summary, the quality of pseudolabels is a key factor that affects the classifier's ability to categorize unlabeled samples. However, pseudolabel learning methods are essentially a self-labeling process, where noise can be amplified during training, ultimately degrading the results.

Unlike pseudolabel learning methods, consistency regularization methods are more oriented toward enhancing model robustness. Consistency regularization is based on the smoothness assumption [33], with the core idea that the model should produce consistent predictions when small perturbations or data augmentations are applied to the same input. Common perturbation strategies include input perturbation, weight perturbation, and network layer perturbation. Sun et al. [34] proposed a class consistency loss, adding spectral constraints and adjusting the attention weights of different branches to integrate spectral and spatial features. Lu et al. [35] introduced a dynamic classifier ensemble objective function, which achieves consistency among multiple classifiers in ensemble learning by minimizing the L2 norm between their outputs, thereby reducing the diversity among

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classifiers. Xu et al. [36] proposed a robust self-ensembling network (RSEN) for hyperspectral classification tasks. This network consists of a baseline network and an ensemble network. Supervised learning is conducted by calculating the classification loss of labeled data on the baseline network and backpropagating the errors. Consistency regularization learning is achieved by minimizing the prediction differences between the baseline and ensemble networks for augmented versions of the unlabeled data. However, due to the complex spatial and spectral heterogeneity of hyperspectral data, simple static augmentation methods gradually become less helpful for capturing the essential spatial–spectral features of the data in the later stages of training.

To address the issues, this article proposes self-ensemble learning based on energy attention and dual-branch guidance (EDSEL) for HSIC. An energy attention-based neural network is designed to extract more discriminative spectral and spatial features from HSI data. The proposed framework comprises student and teacher networks that share the same architecture. The difference lies in the update mechanism: the student network is updated through backpropagation, while the teacher network is updated using the exponential moving average (EMA) method. First, the energy attention unit is obtained by optimizing energy functions computed along pixels' spectral dimensions. Then, these attentions are recalculated and combined via residual connections to preserve multiscale spatial structure information and enhance the discriminability of spectral and spatial features. Second, a teacher-assisted loss (TAL) function is introduced for the supervised learning process on labeled samples. Based on the classification loss of the student network, additional alignment information between the teacher and student networks is incorporated into the backpropagation of the student network. Specifically, the mean squared error (mse) between the feature representations of labeled samples produced by both networks is computed. This encourages the student network to learn knowledge distilled from the teacher network, leading to more separable and stable feature representations for labeled samples. Finally, a dual-branch guidance strategy (DGS) is designed for consistency regularization on unlabeled samples. Two branches apply different data perturbation strategies: one branch uses static noise, while the other introduces temporal noise. The temporal noise starts small at the beginning of training and increases as training progresses. Enforcing consistency between the predictions of the two branches under static and dynamic noise enhances the model's robustness and generalization ability. This strategy can help the network learn the essential spatial–spectral features of land cover objects, ultimately improving the classification accuracy of the unlabeled samples.

The main contributions of this article are given as follows.

1) An energy attention-based self-ensemble network is designed. Spectral and spatial attention mechanisms are optimized using energy functions computed in the spectral and spatial dimensions. The recalculated attentions are combined via residual connections to preserve multiscale spatial structure information and help the deep attention modules obtain fine-grained discriminative information.
2) A TAL function is employed for the learning process of labeled samples. Alignment information between the teacher and student networks is introduced with the classification loss. This facilitates the student network in learning knowledge from the teacher network and enhances its ability to extract more separable and stable features.
3) A DGS is developed for consistency regularization on unlabeled samples. Two branches apply different data augmentation techniques: one uses static noise, and the other uses temporal noise. Consistency regularization between the two branches helps the network learn the essential spatial–spectral features of ground objects.

The remainder of this article is arranged as follows. Section II provides an explicit introduction to EDSEL. Section III presents experiments conducted on three datasets along with the analysis of results. Section IV summarizes this article.

II. Proposed Method

This article proposes EDSEL to address the challenges of limited labeled samples and the diminishing effectiveness of a single perturbation strategy in the self-ensembling learning process of HSIC. The network architecture of the EDSEL is illustrated in Fig. 1.

As shown in Fig. 1, the HSI is first processed by principal component analysis (PCA) to efficiently decrease the number of spectral bands. The resulting low-dimensional data and the original HSI image are carefully divided into labeled and unlabeled samples based on the available number of labels. Each sample contains a pixel vector from the original HSI and a neighborhood pixel patch from the low-dimensional data. An advanced energy attention self-ensemble network (EASN) is then employed to get more discriminative features by optimizing a sophisticated spectral energy function. The EASN comprises a student network and a teacher network, which share the same structure but employ distinct update strategies. The TAL is calculated using the predictions from both networks and the ground-truth labels for the labeled samples. TAL effectively combines alignment information between the student and teacher networks with a classification loss, significantly enhancing its ability to extract more comprehensive features. The DGS is applied to enforce consistency regularization for the unlabeled samples. Specifically, the unlabeled samples are augmented with static and temporal noise, generating static and temporal noisy samples. Similar to the labeled samples, both augmented samples are passed through the student and teacher networks to obtain their respective predictions. The dual-branch loss is computed based on the mse between the predictions of these two branches, along with entropy minimization, to further enhance the robustness and generalization ability of the network. The proposed framework comprises three core components: the EASN (see Section II-A), the TAL (see Section II-B), and the DGS (see Section II-C).

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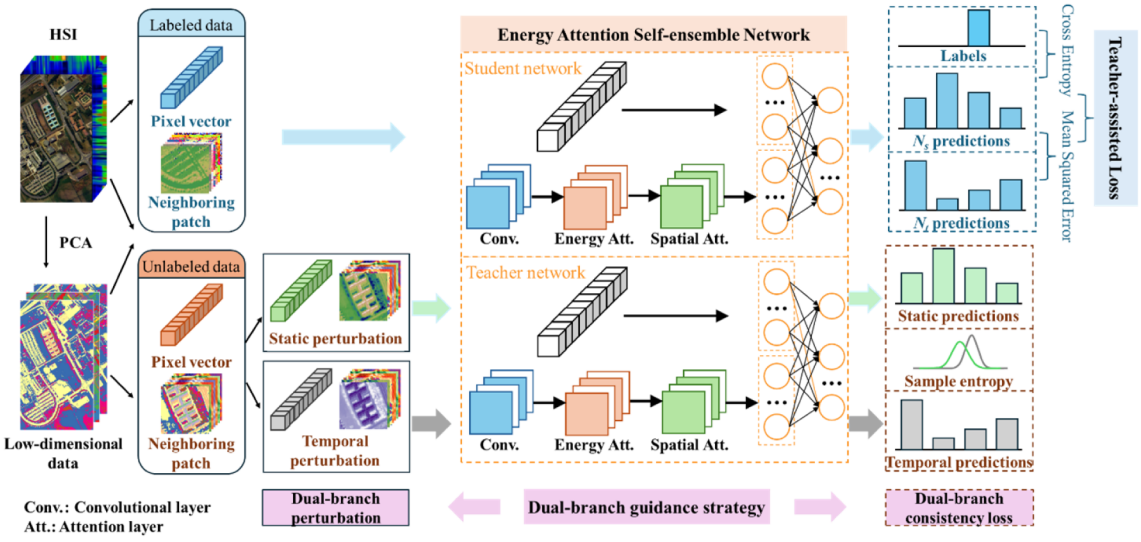


Fig. 1. Overall architecture of the EDSEL network.

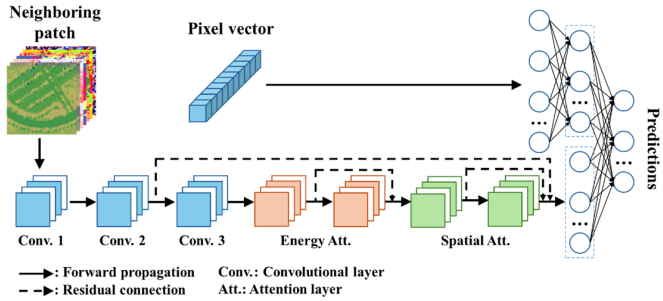


Fig. 2. Framework of the EASN.

A. Energy Attention Self-Ensemble Network

This article proposes EASN to facilitate discriminative and accurate representation for labeled and unlabeled samples in HSI. The network architecture consists of a student network and a teacher network with identical structures, where the detailed architecture of EASN is illustrated in Fig. 2.

As illustrated in Fig. 2, the EASN takes two inputs: the single-pixel vector X_s from the HSI and the neighboring pixel patch X_p from the low-dimensional data. Initially, X_p is passed through three convolutional operations to extract spatial information, which can be expressed as follows:

$$X_1 = F_1\left(X_p\right) \tag{1}$$

$$X_2 = F_p\left(F_a\left(F_2\left(X_1\right)\right)\right) \tag{2}$$

$$X_3 = F_3\left(X_2\right) \tag{3}$$

where F_1, F_2, and F_3 represent three convolutional layers, F_a denotes the ReLU activation function for learning nonlinear features, and F_p indicates the average pooling function. The convolutional kernel size and stride of F_1 are 1×1 and 1, respectively, while F_2 and F_3 employ a 3×3 kernel size with a stride of 1. Subsequently, X_3 is fed into the energy attention unit to capture complex spectral and spatial information. The energy attention unit comprises both spectral energy attention and spatial attention mechanisms.

It is found that in the lateral inhibition phenomenon, activated neurons reduce the activity levels of surrounding neurons by inhibiting them, thereby enhancing their own response to stimuli [37]. This inhibitory mechanism makes the perceptual system more sensitive to edge detection and contrast changes. Inspired by [38], this article applies this mechanism to spectral energy attention. A novel energy function is designed to evaluate the linear separability between the target neuron and other neurons within the same channel. By minimizing this function, a closed-form importance score that quantifies the uniqueness of neurons can be obtained. The score is mapped to attention weights via the sigmoid activation function, thereby achieving adaptive enhancement and suppression of spectral features. First, the input feature is $X \in R^{C\times H\times W}$, where t and x_i denote the target neuron and other neurons of X along the spectral dimension, and its energy function is calculated by measuring the discrepancy between the output of each neuron and the label

$$e_t\left(w_t, b_t, \mathbf{y}, x_i\right) = \left(y_t - \hat{t}\right)^2 + \frac{1}{C-1}\sum_{i=1}^{C-1}\left(y_o - \hat{x}_i\right)^2 \tag{4}$$

where $\hat{t} = w_t t + b_t$ and $\hat{x}_i = w_t x_i + b_t$ represent the outputs of neuron t and its neighboring neuron i, respectively, w_t and b_t are the weight and bias, y_t and y_o denote different labels, and C indicates the total number of neurons. By minimizing the energy function e_t, neuron t can achieve linear separability from its neighboring neurons. Assume that y_t and y_o take values of 1 and -1, respectively, and a regularization term is added to prevent overfitting. Equation (4) can be reformulated as follows:

$$e_t\left(w_t, b_t, \mathbf{y}, x_i\right) = \frac{1}{C-1}\sum_{i=1}^{C-1}\left(w_t x_i + b_t + 1\right)^2 + \\ + \left(1 - \left(w_t t + b_t\right)\right)^2 + \lambda w_t^2. \tag{5}$$

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The partial derivatives of w_t and b_t in (5) are computed as

$$\frac{\partial e_t}{\partial w_t} = w_t\left(2t^2 + 2{\mu_t}^2 + 2{\sigma_t}^2 + 2\lambda\right) + 2\left(t + \mu_t\right)b_t + 2\left(\mu_t - t\right) \tag{6}$$

$$\frac{\partial e_t}{\partial b_t} = 2\mu_t w_t + 2tw_t + 4b_t \tag{7}$$

where μ_t represents the expectation of neighboring neurons and σ_t denotes their variance, which can be calculated using the following formulas:

$$\mu_t = \frac{1}{C-1}\sum_{i=1}^{C-1} x_i, \quad \sigma_t^2 = \frac{1}{C-1}\sum_{i}^{C-1}(x_i - \mu_t)^2. \tag{8}$$

The analytical solutions for w_t and b_t can be obtained by setting (6) and (7) to zero

$$w_t = \frac{2\left(t - \mu_t\right)}{\left(t - \mu_t\right)^2 + 2\sigma_t^2 + 2\lambda} \tag{9}$$

$$b_t = -\frac{\left(t + \mu_t\right)w_t}{2}. \tag{10}$$

Substituting (9) and (10) into (5), the minimum energy obtained is

$$e_t^* = \frac{4\left({\sigma_t}^2 + \lambda\right)}{\left(t - \mu_t\right)^2 + 2{\sigma_t}^2 + 2\lambda}. \tag{11}$$

For computational convenience, it can be assumed that μ_t and σ_t are equal to all neurons' average and variance, respectively. By substituting (9) and (10) into (5), the minimum energy can be obtained as follows:

$$e_t^* = \frac{4\left(\hat{\sigma}^2 + \lambda\right)}{\left(t - \hat{\mu}\right)^2 + 2\hat{\sigma}^2 + 2\lambda} \tag{12}$$

where $\hat{\mu}$ and $\hat{\sigma}^2$ represent all neurons' average and variance. In terms of computational complexity, (11) needs to be computed separately for each channel, while (12) only requires calculating the mean and variance once globally, making it faster than (11). Although using $\hat{\mu}$ and $\hat{\sigma}^2$ will lose local subtle differences, it can reduce the impact caused by abnormal neurons and make the attention weights more stable. According to (12), the greater the deviation of neuron t from the $\hat{\mu}$, the lower its energy. Therefore, the importance of each neuron can be determined by

$$\frac{1}{e_t^*} = \frac{\left(t - \hat{\mu}\right)^2}{4\left(\hat{\sigma}^2 + \lambda\right)} + \frac{1}{2}. \tag{13}$$

By applying the importance function to the spectral dimension, the importance of the image's spectral dimension can be calculated as follows:

$$I\left(X_3\right) = F_b\left(\frac{\left(X_3 - \mu_S\left(X_3\right)\right)^2}{4\left(\sigma_S\left(X_3\right)^2 + \lambda\right)} + \frac{1}{2}\right) \tag{14}$$

where μ_S and σ_S denote the expectation and variance of X_3 along the spectral dimension, and F_b represents the sigmoid activation function. The output of the spectral energy attention unit U_S is given by

$$U_S = X_3 \cdot I\left(X_3\right). \tag{15}$$

The output of the spectral energy attention block is obtained after reusing the U_S. Specifically, U_S is added to its residual version, which can be expressed as

$$X_4 = U_S + U_S \cdot I\left(U_S\right). \tag{16}$$

By combining the output of the spectral attention unit with its residual version, the transfer ability of valuable features can be enhanced, further capturing the dependence of spectral dimensions and effectively alleviating the overfitting problem in the model. Similarly, the result of the spatial attention block is obtained by combining the output of the spatial attention unit with its residual version. The spatial attention unit adopts a cross-attention mechanism, which enhances the expression ability of features and reduces computation cost by calculating the attention weights of feature maps in the horizontal and vertical directions. The output of the spatial attention unit U_C is represented as follows:

$$U_C = F_C\left(X_4\right) \tag{17}$$

where F_C is a cross-attention network. Specifically, $X_4 \in R^{B\times C\times H\times W}$, where C is the number of channels of the feature, H is the height, and W is the width. First, three features $Q \in R^{B\times C'\times H\times W}$, $K \in R^{B\times C'\times H\times W}$, and $V \in R^{B\times C\times H\times W}$ are obtained via convolution

$$Q = W_q * X_4, \mathrm{K} = W_k * X_4, \mathrm{V} = W_v * X_4 \tag{18}$$

where C' is the reduced number of channels, and W_q, W_k, and W_v denote three 1×1 convolutions. Subsequently, the features Q, K, and V are reshaped along the height and width dimensions

$$\begin{aligned} Q_H &= RS_H\left(Q\right), \quad Q_W = RS_W\left(Q\right) \\ K_H &= RS_H\left(K\right), \quad K_W = RS_W\left(K\right) \\ V_H &= RS_H\left(V\right), \quad V_W = RS_W\left(V\right) \end{aligned} \tag{19}$$

where RS_H and RS_W are dimension adjustment functions for the height and width dimensions, $Q_H \in R^{BW\times H\times C'}$, $Q_W \in R^{BH\times W\times C'}$, $K_H \in R^{BW\times C'\times H}$, $K_W \in R^{BH\times C'\times W}$, $V_H \in R^{BW\times C\times H}$, and $V_W \in R^{BH\times C\times W}$. Subsequently, the similarity in the height and width dimensions is calculated separately

$$\begin{aligned} E_H = Q_H \cdot K_H, E_W = Q_W \cdot K_W \\ A = \mathrm{Softmax}\left(\mathrm{Concat}\left(E_H, E_W\right)\right) \\ A_H, A_W = Split\left(A\right) \end{aligned} \tag{20}$$

where *Concat* denotes the concatenation operation, *Softmax* is the normalization function, $A \in R^{B\times H\times W\times(H+W)}$, and *Split* is the splitting function that divides A into $A_H \in R^{B\times H\times W\times H}$ and $A_W \in R^{B\times H\times W\times W}$. Finally, the output of the spatial attention unit is obtained as

$$\begin{aligned} \mathrm{O}_H = \mathrm{V}_\mathrm{H} \cdot A_H^T, \mathrm{O}_W = \mathrm{V}_\mathrm{W} \cdot A_W^T \\ U_C = X_4 + \mathrm{O}_H + \mathrm{O}_W. \end{aligned} \tag{21}$$

The result of the spatial attention block is calculated as follows:

$$X_5 = U_C + F_C\left(U_C\right). \tag{22}$$

The different layer results X_2 and X_5 are connected via a residual connection to avoid the gradient problem of EASN

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and followed by activation and pooling operations to obtain a more discriminative feature representation X_6:

$$X_6 = F_p\left(F_a\left(X_2 + X_5\right)\right). \tag{23}$$

Next, the discriminative feature X_6 is concatenated with the central pixel vector X_s to enhance the model's learning of the central pixel. X_s is processed to match the same dimension as X_6 for the concatenation operation

$$X_A = cat\left(X_6, C_1\left(X_s\right)\right) \tag{24}$$

where *cat* represents the concatenation operation, which merges feature maps from different depths to enhance the model's learning of the central pixel features. C_1 is a fully connected layer used to transform X_s to the same dimension as X_6. Then, the concatenated feature X_A is passed through a fully connected layer for comprehensive learning, further enhancing the model's representational power

$$X_F = F_a\left(C_2\left(X_A\right)\right) \tag{25}$$

where C_2 is a fully connected layer and F_a is the ReLU activation function. Finally, the classifier generates the prediction results P for the image

$$P = C_3\left(X_F\right) \tag{26}$$

where C_3 is a fully connected layer and the output dimension of C_3 equals the number of classes in the HSI.

EASN obtains the initial attention features U_S and U_C through the spectral and spatial attention units. Then, EASN combines U_S and U_C with the corresponding residual version to fully use the domain information to obtain the total attention features, alleviating the network's overfitting phenomenon.

B. Teacher-Assisted Loss

This article designs a TAL in the supervised learning process of labeled samples. Built upon the standard classification loss, TAL incorporates alignment information between the teacher and student networks to guide the student's backpropagation. This mechanism helps the student network learn knowledge from the teacher network, enhancing its ability to extract discriminative features and obtain a stable representation of labeled samples.

The inputs X_s^l and X_p^l are passed to the student network N_s to obtain the corresponding category prediction results y_s^l

$$y_s^l = N_s\left(X_s^l, X_p^l\right). \tag{27}$$

The classification loss of labeled samples can be calculated based on the predicted results y_s^l and the labels $\hat{y}^l$

$$L_c^l = CE\left(y_s^l, \hat{y}^l\right) \tag{28}$$

where CE is the cross-entropy loss function. The similarity loss of labeled samples is calculated by mse between the predicted results in the student and teacher networks

$$L_s^l = \frac{1}{n_l}\left\|y_s^l - y_t^l\right\|_2^2 \tag{29}$$

where n_l is the number of labeled samples, and y_s^l and y_t^l are the prediction results of student network N_s and teacher network N_t for labeled samples, respectively. The calculation formula is given as follows:

$$y_t^l = N_t\left(X_s^l, X_p^l\right). \tag{30}$$

The proposed TAL is calculated as follows:

$$TAL = L_c^l + \lambda_s L_s^l \tag{31}$$

where λ_s is the weighting factor. The regularization computed by the similarity loss is introduced to the student's classification loss. Since the teacher network incorporates information from unlabeled samples, as discussed in Section II-C, the similarity loss for labeled samples allows the student network to leverage information from unlabeled data, enhancing its ability to learn the overall data distribution.

C. Dual-Branch Guidance Strategy

Self-ensemble learning leverages consistency regularization to fully utilize unlabeled samples, enhancing model robustness and generalization. Specifically, the core idea of consistency regularization is to apply perturbations to unlabeled samples and train the model to maintain consistent outputs under these perturbations. These perturbations can include noise interference, adversarial perturbation, and data augmentation. As gradient stability increases in the later stage of network training, the effect of static noise perturbations diminishes. Therefore, this article proposes a DGS, where two branches employ static and temporal noise for data augmentation of unlabeled samples. The temporal noise gradually increases as training progresses, starting with a low value at the beginning and adopting a higher value toward the end of training, facilitating learning in the later stage.

First, two types of noise perturbations are added to the unlabeled sample inputs X_s^u and X_p^u

$$\begin{aligned} X_1^u &= P_S\left(X_s^u, X_p^u\right) \\ X_2^u &= P_T\left(X_s^u, X_p^u\right) \end{aligned} \tag{32}$$

where P_S and P_T are static noise perturbations and temporal noise perturbations, and X_1^u and X_2^u are outputs corresponding to P_S and P_T. The calculation progress of P_S and P_T is given as follows:

$$\begin{aligned} P_S\left(x\right) &= x + 0.5zx \\ P_T\left(x\right) &= x + \tanh\left(\tanh^{-1}\left(0.99\right)\frac{i}{I}\right)zx \end{aligned} \tag{33}$$

where $z \sim \mathcal{N}(0,1)$, I denotes the total number of training iterations, and i denotes the current training iteration. As i varies from 0 to I, the value of $\tanh(\tanh^{-1}(0.99)i/I)$ increases monotonically from 0 to 0.99, ensuring a continuous increase in perturbation during training. Meanwhile, the perturbation grows rapidly in the early stage of training, which quickly strengthens the perturbation intensity and accelerates convergence. In the later training stage, the growth of perturbation slows down, ensuring a stable and nonoscillatory training process, while maintaining an appropriate consistency constraint to refine the generalization ability of the model. The $tanh(x)$ function has a gradient saturation region, that is, when x is sufficiently large, the output changes only slightly.

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To avoid vanishing gradients during backpropagation, this article adopts $\tanh^{-1}(0.99)$ for truncation, which helps keep the gradient normal throughout the training process and provides a sufficient upper bound for the perturbation to facilitate network training. Then, the two types of perturbations are replicated for the consistency regularization, and the replicated results are fed into the N_s and N_t to obtain outputs

$$y_{s1}^u = \frac{1}{m} N_s\left(rep\left(X_1^u, m\right)\right), \quad y_{s2}^u = \frac{1}{m} N_s\left(rep\left(X_2^u, m\right)\right) \tag{34}$$

$$y_{t1}^u = \frac{1}{m} N_t\left(rep\left(X_1^u, m\right)\right), \quad y_{t2}^u = \frac{1}{m} N_t\left(rep\left(X_2^u, m\right)\right) \tag{35}$$

where m is the number of replications and rep represents the replication operation. y_{s1}^u, y_{s2}^u, y_{t1}^u, and y_{t2}^u are the prediction results of the samples processed by P_S and P_T through the N_s and N_t. The consistency loss for the samples under the effect of P_S is calculated as follows:

$$L_{PS} = \frac{1}{n_u} \left\| y_{s1}^u - y_{t1}^u \right\|_2^2 \tag{36}$$

where n_u represents the number of unlabeled samples. The consistency loss under the P_T is calculated as follows:

$$L_{PT} = \frac{1}{n_u} \left\| y_{s2}^u - y_{t2}^u \right\|_2^2 \tag{37}$$

where n_u represents the number of unlabeled samples. The consistency loss for dual-branch perturbations is calculated as follows:

$$L_D = L_{PS} + L_{PT} + \frac{1}{n_u} \left\| y_{s1}^u - y_{s2}^u \right\|_2^2 + \lambda_H \left(H\left(y_{s1}^u\right) + H\left(y_{s2}^u\right)\right) \tag{38}$$

where $H(x)$ represents the entropy of distribution x and λ_H is the weight of the entropy of the predicted results of unlabeled samples under P_S and P_T in N_s. In L_D, L_{PS} and L_{PT} focus on the consistency of single perturbation across models, using the teacher network to supervise the student network. The third term focuses on the cross-perturbation consistency of the student model, ensuring that the student model produces consistent predictions under different perturbations, so that the model concentrates on the essential spectral–spatial characteristics of ground objects. Entropy regularization prevents the student model from being overconfident in unlabeled samples, facilitates learning the diversity of unlabeled samples, and effectively addresses the core challenge of spectral heterogeneity in hyperspectral data.

The overall loss of the network is given as follows:

$$L = TAL + \lambda_D L_D \tag{39}$$

where TAL is the loss function for labeled samples, L_D is the loss function for unlabeled samples, and λ_D is the weight.

After obtaining the overall loss, N_s updates its weight using the gradient backpropagation method, while N_t updates its weights using the EMA method [39]

$$W_t^i = \alpha W_t^{i-1} + (1-\alpha) W_s^i \tag{40}$$

where W_t^{i-1} represents the N_t's weights from the previous training iteration, W_s^i represents the N_s's weights from the current training iteration, and α is the smoothing factor. The EMA-based update allows N_t to provide stable prediction targets while remaining sensitive to changes in the N_s's weight during training. This enhances the learning capability for unlabeled samples and improves the overall classification performance of the network.

Algorithm 1 EDSEL: Self-Ensemble Learning Based on Energy Attention and Dual-Branch Guidance

Input: HSI data X, student network N_s, teacher network N_t, the number of iterations N.
Output: The classification result Y.

```
Obtain pixel vectors X_s and neighborhood pixel patch X_P of X by PCA;
Divide X_s and X_P into labeled samples and unlabeled samples based on the number of labels;
for i = 1, 2, ..., N do
    if labeled samples then
        Calculate the predicted results y_s^l and y_t^l for the N_s and N_t using Eq. (27) and (30);
        Get TAL using Eq. (31);
    else
        Obtain noise perturbation samples X_1^u and X_2^u using Eq. (32);
        Calculate the prediction results y_s1^u and y_s2^u of the N_s for X_1^u and X_2^u using Eq. (34);
        Calculate the prediction results y_t1^u and y_t2^u of the N_t for X_1^u and X_2^u using Eq. (35);
        Calculate the consistency loss L_D using Eq.(38);
    end if
    Calculate the consistency loss L using Eq.(39);
    if N_s then
        Update network weights based on backpropagation;
    else
        Update network weights using Eq. (40);
    end if
end for
Use N_s to classify unlabeled samples and obtain classification results Y.
return: Y.
```

The basic process of the proposed EDSEL algorithm is provided in Algorithm 1.

III. Experiment Results and Analysis

A. Datasets Description

This article conducts experiments on three datasets widely used in HSIC to verify the effectiveness of the proposed method. These three datasets are Pavia University (PU), Salinas (SA), and Houston (HU). The detailed introduction of the three datasets is given as follows.

1) *PU Dataset:* Acquired by the reflective optics system imaging spectrometer over the University of Pavia in Italy, this dataset consists of a 610 × 340-pixel image with 115 spectral bands spanning 0.43–0.86 μm. After removing 12 noisy bands, 103 spectral bands remain. The dataset encompasses nine land cover types, including asphalt, grass, gravel, trees, metal sheets, bare soil,

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TABLE I
COLOR, LAND COVER, AND NUMBER OF SAMPLES FOR THE PU DATASET

Class	Color	Land cover class	Samples
0		Background	164624
1		Asphalt	6631
2		Meadows	18649
3		Gravel	2099
4		Trees	3064
5		Painted metal sheets	1345
6		Bare Soil	5029
7		Bitumen	1330
8		Self-Blocking Bricks	3682
9		Shadows	947
		Total	207400

TABLE II
COLOR, LAND COVER, AND NUMBER OF SAMPLES FOR THE SA DATASET

Class	Color	Land cover class	Samples
0		Background	10895
1		Broccoli green weeds 1	2009
2		Broccoli green weeds 2	3726
3		Fallow	1976
4		Fallow rough plow	1394
5		Fallow smooth	2678
6		Stubble	3959
7		Celery	3579
8		Grapes untrained	11271
9		Soil vineyard develop	6203
10		Corn senesced green weeds	3278
11		Lettuce romaine 4wk	1068
12		Lettuce romaine 5wk	1927
13		Lettuce romaine 6wk	916
14		Lettuce romaine 7wk	1070
15		Vinyard untrained	7268
16		Vinyard vertical trellis	1807
		Total	65024

bitumen, bricks, and shadows, totaling 42 776 labeled pixels. Table I provides a detailed overview of each class's colors, land cover information, and sample numbers in the PU dataset.

2) *SA Dataset:* Collected by the Airborne Visible/Infra-Red Imaging Spectrometer over Salinas Valley, California, this dataset covers an area of 512 × 217 pixels. After removing spectral bands affected by water absorption, the total number of spectral bands is reduced from 224 to 220. It comprises 16 land cover types, including vegetables, bare soil, and vineyards. Table II provides a detailed overview of each class's colors, land cover information, and sample numbers in the SA dataset.
3) *HU Dataset:* Collected by the National Center for Airborne Laser Mapping, this dataset covers the University of Houston and its surrounding areas in Texas, USA. It comprises 349 × 1905 pixels with 144 spectral bands, spanning a wavelength range of 0.38–1.04 μm. The dataset features 15 land cover types, including grass, parking lots, and runways, with 53 649 labeled pixels. Table III provides a detailed overview of each class's colors, land cover information, and sample numbers in the HU dataset.

TABLE III
COLOR, LAND COVER, AND NUMBER OF SAMPLES FOR THE HU DATASET

Class	Color	Land cover class	Samples
0		Background	611196
1		Healthy grass	1251
2		Stressed grass	1254
3		Synthetic grass	697
4		Trees	1244
5		Soil	1252
6		Water	325
7		Residential	1268
8		Commercial	1244
9		Road	1252
10		Highway	1227
11		Railway	1235
12		Parking Lot 1	1233
13		Parking Lot 2	469
14		Tennis Court	428
15		Running Track	660
		Total	664845

B. Comparison Algorithms and Experimental Settings

Experiments are conducted with six state-of-the-art algorithms to validate the effectiveness of the proposed method. These algorithms include deep contextual CNN (ContextNet) [40], deep pyramidal residual network (DPyResNet) [41], spectral–spatial residual network (SSRN) [42], attention-based adaptive spectral–spatial kernel ResNet (A2S2KNet) [43], multilevel prototype alignment network (MLPA) [44], RSEN [36], and spatial–spectral enhancement and fusion network (SSEFN) [45]. RSEN is an SSL algorithm, while the others are supervised learning algorithms. In the experimental setup, 30 samples per class are selected as training samples across all three datasets, with the remaining samples used for consistency regularization learning. After PCA, the number of spectral bands is reduced to 5. The patch size is 16 × 16, the number of replication m in (29) is 5, and the λ_H in (33) is set to 0.0001. The network is optimized using the Adam optimizer with a learning rate of 0.0005. All experiments use the PyTorch framework, running on CPU Xeon E5-2630 and GPU an NVIDIA Titan X. Each experiment is repeated ten times to prevent randomness from affecting the results.

C. Evaluation Metrics

This article evaluates the classification results of six state-of-the-art algorithms and EDSEL using three metrics: overall accuracy (OA), average accuracy (AA), and kappa coefficient (KC). OA represents the proportion of correctly classified samples to the total samples. AA is the AA across all classes, indicating the model's ability to maintain balanced performance across different categories. KC is particularly useful for assessing classification consistency in the presence of class imbalance, providing a more comprehensive evaluation under uneven category distributions.

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TABLE IV
CLASSIFICATION RESULTS ON THE SA DATASET (%, 30 LABELED SAMPLES PER CLASS)

Class	ContextNet	DPyResNet	SSRN	A2S2KNet	MLPA	RSEN	SSEFN	EDSEL
1	94.69±3.99	93.54±11.61	99.71±0.45	96.33±0.99	99.52±0.74	99.51±0.53	**99.86±0.09**	99.71±0.15
2	98.94±1.33	99.25±1.12	98.43±3.12	98.81±0.27	99.83±0.29	99.34±1.05	99.82±0.28	**99.86±0.41**
3	97.51±2.33	96.06±2.39	98.40±1.41	98.14±1.95	99.70±0.56	99.47±1.38	99.44±0.48	**99.89±0.14**
4	88.63±8.82	92.14±6.77	96.76±3.57	97.70±1.72	99.06±1.45	99.46±0.45	98.82±0.24	**99.88±0.05**
5	92.15±6.16	97.88±1.36	98.81±2.08	**99.74±0.19**	97.49±1.12	99.29±0.46	98.04±1.18	99.09±0.24
6	97.54±4.10	99.45±0.79	99.74±0.45	99.65±0.63	**100.00±0.00**	**100.00±0.00**	99.67±0.32	**100.00±0.00**
7	99.82±0.29	99.91±0.14	**99.99±0.03**	99.74±0.40	99.50±0.81	99.73±0.29	99.97±0.05	**99.99±0.02**
8	70.60±6.32	85.37±8.52	78.17±11.21	88.57±6.54	85.02±3.82	79.82±5.21	**90.63±2.20**	85.32±4.26
9	98.90±1.55	98.88±2.26	99.77±0.23	98.52±2.14	99.82±0.25	99.80±0.25	99.44±0.16	**99.91±0.07**
10	93.99±5.02	87.15±11.04	94.77±5.18	95.92±2.67	**99.62±2.10**	97.71±1.07	96.30±1.31	98.16±0.54
11	87.75±6.47	93.47±4.74	96.66±2.55	96.23±3.97	99.65±0.41	99.38±0.63	95.56±2.83	**99.78±0.08**
12	98.12±2.50	98.68±1.24	99.68±0.26	99.59±0.89	99.87±0.29	99.96±0.15	99.75±0.28	**100.00±0.00**
13	97.55±4.97	92.97±12.52	99.73±0.46	99.79±0.57	99.50±0.37	99.93±0.16	99.91±0.10	**100.00±0.00**
14	98.09±2.71	97.65±2.63	98.46±1.44	97.18±3.36	99.35±0.53	**99.55±0.62**	97.77±0.74	99.43±0.25
15	53.86±6.84	57.47±6.38	61.88±11.88	58.27±13.33	86.62±3.70	91.56±3.55	78.44±3.93	**92.27±2.81**
16	93.46±10.61	99.50±0.88	**99.98±0.05**	99.16±1.62	99.24±0.60	98.04±2.50	99.93±0.15	99.32±0.19
OA	84.95±2.35	86.44±2.78	87.34±2.95	87.49±2.18	94.15±0.83	94.22±1.74	94.30±0.71	**95.65±1.69**
AA	91.35±1.40	93.08±1.10	95.06±0.73	95.10±0.17	97.55±0.37	97.64±1.19	97.09±0.34	**98.29±0.46**
KC	83.29±2.56	85.00±3.01	85.93±3.20	86.19±2.59	93.45±0.92	93.58±0.74	93.65±0.78	**95.16±2.17**

TABLE V
CLASSIFICATION RESULTS ON THE HU DATASET (%, 30 LABELED SAMPLES PER CLASS)

Class	ContextNet	DPyResNet	SSRN	A2S2KNet	MLPA	RSEN	SSEFN	EDSEL
1	87.76±5.08	88.05±4.90	87.39±6.55	92.49±4.82	**98.17±0.89**	97.63±0.23	94.04±3.72	97.99±0.43
2	88.87±3.86	91.86±5.98	94.26±9.74	96.50±3.58	93.76±4.61	98.65±0.75	93.20±3.95	**99.95±0.07**
3	86.77±5.49	96.66±4.99	99.95±0.14	99.65±0.60	99.53±0.47	99.96±0.10	**100.00±0.00**	**100.00±0.00**
4	86.26±12.88	92.48±7.75	94.76±3.30	96.26±2.16	97.65±2.96	97.77±0.38	98.36±1.69	**98.88±0.52**
5	95.44±2.56	92.43±5.99	95.91±3.82	94.68±4.68	99.61±0.35	99.67±0.37	96.18±2.02	**100.00±0.00**
6	98.83±2.15	88.44±9.31	89.80±13.93	93.97±3.97	96.44±2.46	**100.00±0.00**	99.93±0.21	**100.00±0.00**
7	91.14±11.00	88.65±8.14	91.59±4.90	94.46±3.12	90.40±4.21	86.63±0.86	**94.85±2.71**	86.37±1.17
8	94.68±5.14	94.74±2.73	96.73±4.79	**96.90±3.03**	81.31±4.41	80.68±1.51	96.82±2.70	81.92±0.80
9	87.12±16.84	83.35±7.45	85.35±7.81	92.16±2.21	85.93±2.76	89.24±1.40	90.69±4.09	**93.67±1.57**
10	82.43±8.17	81.14±7.86	75.19±10.93	83.19±2.75	95.56±2.96	97.94±0.66	87.18±5.21	**99.97±0.04**
11	89.01±9.94	90.35±8.40	90.78±10.69	**97.54±1.71**	90.29±4.09	90.27±1.23	96.09±3.74	95.44±1.59
12	78.97±9.64	86.68±6.49	81.80±9.23	88.98±7.31	94.00±3.64	90.70±2.45	91.60±3.64	**97.99±2.94**
13	92.18±4.27	86.52±8.37	88.84±8.55	94.49±2.60	**97.58±1.66**	87.04±3.86	93.12±2.64	84.53±5.15
14	87.69±3.49	96.39±6.13	94.59±6.32	96.32±3.78	99.75±0.39	99.70±0.42	97.07±2.32	**100.00±0.00**
15	83.34±7.84	90.93±7.88	94.98±7.87	96.37±1.88	99.84±0.27	99.83±0.33	98.42±1.51	**100.00±0.00**
OA	85.50±8.44	88.57±2.29	89.17±3.24	93.51±1.37	93.70±0.96	93.68±0.31	94.30±1.20	**95.53±0.46**
AA	87.76±5.08	89.91±1.86	90.80±2.91	94.26±1.02	94.66±0.89	94.38±0.30	95.17±0.88	**95.78±0.51**
KC	88.87±3.86	87.65±2.48	88.29±3.51	92.98±1.48	93.19±1.04	93.17±0.33	93.83±1.30	**95.17±0.50**

D. Classification Results and Analysis

The results of six state-of-the-art algorithms and EDSEL on the SA dataset are shown in Table IV.

According to the experimental results on the SA dataset presented in Table IV, EDSEL achieves the best performance on OA, AA, and KC when the number of training samples is set to 30 per class. EDSEL outperforms by 1.43%, 0.65%, and 1.58% compared to the suboptimal method in these three metrics, respectively. From the perspective of OA, the results of the SSL-based RSEN and EDSEL methods are significantly higher than those of supervised learning-based methods. This is because SSL enhances network robustness by enforcing consistency regularization. Moreover, EDSEL surpasses RSEN in performance, indicating that the TAL enhances the network's ability to learn discriminative features of samples.

The results of six state-of-the-art algorithms and EDSEL on the HU dataset are shown in Table V.

Table V shows that when the number of training samples is 30 per class, the EDSEL algorithm achieves the best results in OA, AA, and KC, surpassing the suboptimal results by 1.83%, 0.61%, and 1.34%, respectively. EDSEL achieves the highest classification accuracy in ten out of 15 classes in the HU dataset and attains 100% classification accuracy in five classes, including synthetic grass, soil, and water. This is because the EDSEL algorithm adopts DGS to provide progressively enhanced constraints during training through static and temporal perturbations. Meanwhile, the cross-perturbation consistency loss and entropy regularization can prevent the model from being overconfident and effectively address the core challenge of spectral heterogeneity in hyperspectral data.

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TABLE VI
CLASSIFICATION RESULTS ON THE PU DATASET (%, 30 LABELED SAMPLES PER CLASS)

Class	ContextNet	DPyResNet	SSRN	A2S2KNet	MLPA	RSEN	SSEFN	EDSEL
1	94.22±2.20	89.87±4.71	**98.02±3.07**	99.28±0.45	90.73±2.24	94.15±3.12	98.82±1.84	98.95±0.31
2	97.17±0.95	94.58±2.16	98.01±1.27	97.79±0.99	89.55±3.28	98.19±3.98	**99.22±0.25**	96.91±4.53
3	62.06±7.30	64.93±15.01	92.16±5.27	96.07±3.52	83.48±4.42	91.29±3.91	76.07±7.46	**97.06±1.10**
4	89.50±5.35	91.26±5.54	87.75±11.52	93.28±7.99	95.44±1.71	98.57±0.64	95.20±4.89	**99.44±0.16**
5	97.44±1.63	98.07±1.89	99.41±0.47	99.12±0.92	99.83±0.21	99.33±0.89	**100.00±0.00**	**100.00±0.00**
6	79.01±7.89	70.96±13.11	68.93±11.66	80.91±7.89	**91.83±4.09**	80.82±5.87	86.25±6.16	88.41±2.72
7	59.97±6.96	57.47±11.02	76.82±15.02	80.69±6.65	90.65±3.58	91.29±4.11	87.50±4.36	**92.65±1.20**
8	80.10±5.05	79.81±4.72	89.94±3.66	94.09±1.48	88.29±3.85	93.34±4.33	88.41±3.22	**94.52±4.82**
9	77.82±15.81	82.15±15.73	97.53±2.07	97.66±1.41	99.47±0.42	99.47±0.95	98.12±2.78	**99.98±0.07**
OA	87.37±2.21	84.23±2.86	89.69±2.78	94.10±1.81	90.58±1.49	94.65±2.20	94.27±1.46	**96.24±2.24**
AA	81.92±2.63	81.01±3.12	89.84±2.47	93.21±1.89	92.14±0.66	92.87±0.57	92.18±1.68	**95.03±0.94**
KC	83.53±2.78	79.41±3.53	86.72±3.44	92.27±2.31	87.70±1.88	94.05±2.79	92.48±1.88	**96.43±2.55**

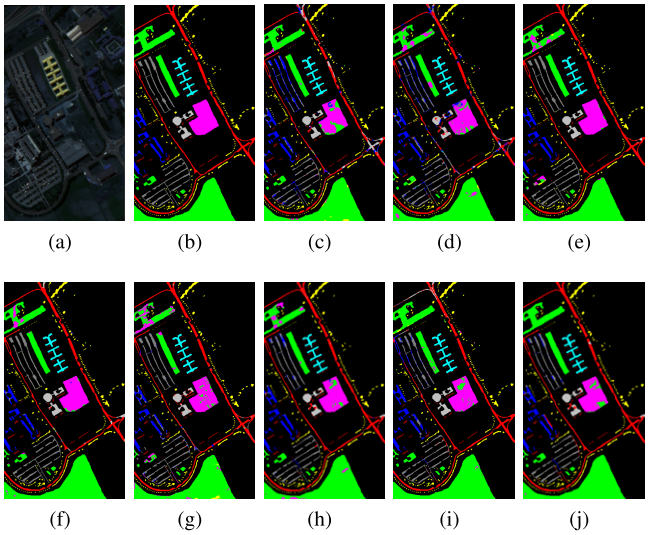


Fig. 3. Visualization maps of various algorithms on the PU dataset. (a) False-color image. (b) Ground truth. (c) ContextNet. (d) DPyResNet. (e) SSRN. (f) A2S2KNet. (g) MLPA. (h) RSEN. (i) SSFEN. (j) EDSEL.

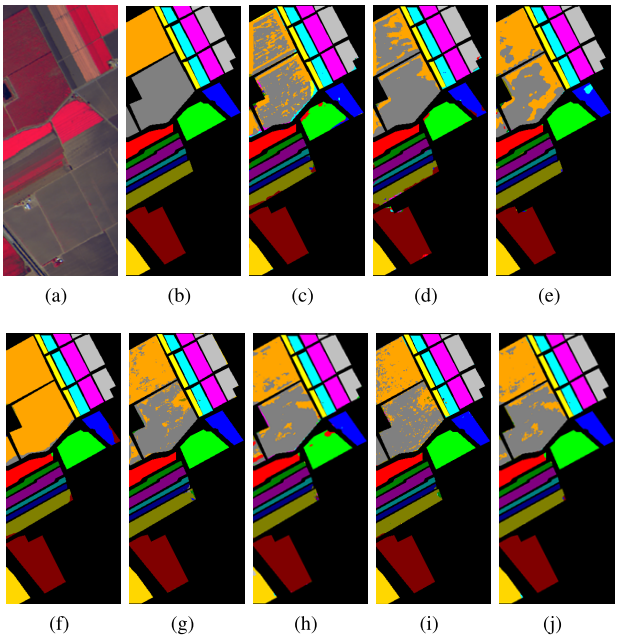


Fig. 4. Visualization maps of various algorithms on the SA dataset. (a) False-color image. (b) Ground truth. (c) ContextNet. (d) DPyResNet. (e) SSRN. (f) A2S2KNet. (g) MLPA. (h) RSEN. (i) SSFEN. (j) EDSEL.

The results of six state-of-the-art algorithms and EDSEL on the PU dataset are shown in Table VI.

As shown in Table VI, EDSEL achieves the best results on OA, AA, and KC. Compared with the suboptimal result, EDSEL outperforms by 1.59%, 2.16%, and 2.38% on OA, AA, and KC, respectively. This is because the EDSEL algorithm adopts EASN to extract discriminative features of HSIs and models the complex band correlations and spatial dependencies in hyperspectral data with the help of spectral energy attention and spatial attention. Meanwhile, residual connections assist deep attention modules in learning fine-grained discriminative information and improve the ability of the network to distinguish different samples. In Table VI, the EDSEL algorithm achieves the best results on six classes.

The visualization results of six state-of-the-art algorithms and EDSEL on the PU dataset are shown in Fig. 3.

As shown in Fig. 3(b), test samples are represented by pixels in different colors, while uninvolved samples are depicted as black pixels in the background. As shown in Fig. 3(b), most classification result maps contain significant noise in the sixth class, particularly in the central region of the image. Referring to Table I, Fig. 3(c) shows that the ContextNet algorithm misclassifies first-class (asphalt) samples, which may be due to its poor learning performance on the narrowly distributed first-class (asphalt) samples. Fig. 3(d)–(h) indicates that the five algorithms DPyResNet, SSRN, A2S2KNet, MLPA, and RSEN misclassify second class (meadows) samples in the top-left corner of the PU scene, and DPyResNet, MLPA, and RSEN also misclassify second class (meadows) samples at the bottom of the PU scene. In Fig. 3(i), the SSFEN algorithm misclassifies the slenderly distributed eighth class (self-blocking bricks) samples as fourth class (trees) samples. The visual classification result of EDSEL in Fig. 3(j) is generally better, which verifies that EDSEL can effectively model the complex spatial dependencies in hyperspectral data.

The visualization results of six state-of-the-art algorithms and EDSEL on the SA dataset are shown in Fig. 4.

From Fig. 4(c), (d), and (h), it is evident that in addition to misclassifying the eighth class, ContextNet, DPyResNet, and

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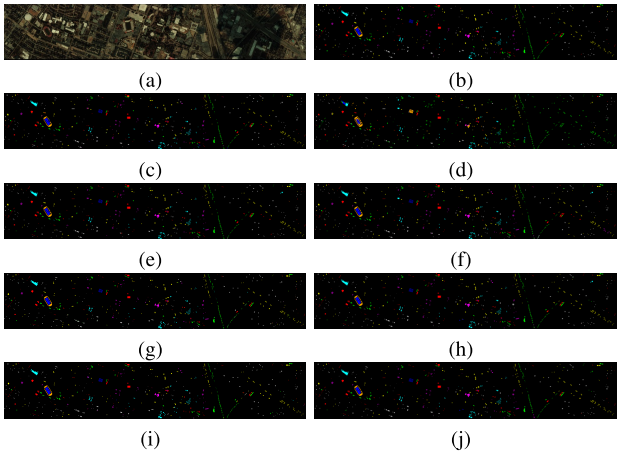


Fig. 5. Visualization maps of various algorithms on the HU dataset. (a) False-color image. (b) Ground truth. (c) ContextNet. (d) DPyResNet. (e) SSRN. (f) A2S2KNet. (g) MLPA. (h) RSEN. (i) SSFEN. (j) EDSEL.

RSEN also misclassify some pixels of the second class (broccoli green weeds 2) as the first class (broccoli green weeds 1). Fig. 4(e) shows that the SSRN algorithm misclassifies some pixels of the third class (fallow) as the fifth class (fallow smooth). Fig. 4(f), and (g) further reveals that the A2S2KNet and MLPA algorithms exhibit significant classification errors when encountering the eighth class (grapes untrained) in the upper-left corner of the image. According to Table II, both A2S2KNet and MLPA misclassify most pixels of the eighth class as the 15th class (vineyard untrained). The SSEFN algorithm produces some speckle-like errors in the upper-left corner of Fig. 4(i). Fig. 4(j) indicates that while the EDSEL algorithm also makes errors in distinguishing between the eighth class and the 15th class, it achieves a higher level of correct classification for other classes.

The visualization results of six state-of-the-art algorithms and EDSEL on the HU dataset are shown in Fig. 5.

As shown in Fig. 5, the HU dataset has a lower proportion of labeled samples than the other two datasets, resulting in most background pixels in the classification maps. Fig. 5(d) shows that DPyResNet misclassifies the first class (healthy grass) on the right-hand side of the image. Fig. 5(c), (e), and (f) illustrates that ContextNet, SSRN, and A2S2KNet incorrectly classify the elongated fourth class (trees) as the first class (healthy grass) on the right-hand side of the image. In the middle-left region of Fig. 5(f), A2S2KNet misclassifies the 12th class (parking lot 1) as the fifth class (soil), possibly due to its limited ability to capture global contextual information. Fig. 5(g) and (h) exhibits minor classification errors. The SSEFN algorithm produces misclassifications in some boundary regions in Fig. 5(i). The classification map in Fig. 5(j) is closest to that in Fig. 5(b), demonstrating that the consistency loss under different perturbations can help the model focus on the essential spectral–spatial characteristics of ground objects.

E. Parameter Settings

In (39) and (31), λ_D and λ_s control the importance of the dual-branch consistency loss and TAL, respectively. Experiments are conducted with λ_D and λ_s selected from [0.1, 0.5, 1, 5, 10] to determine the optimal values. The experimental results on the PU, SA, and HU datasets are shown in Fig. 6.

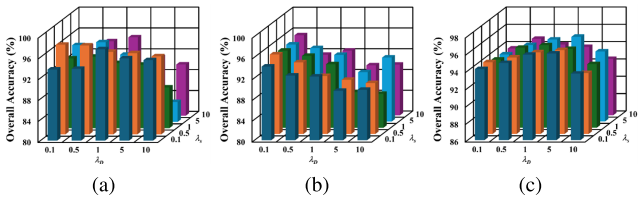


Fig. 6. Experimental results of different λ_D's and λ_s's on three datasets. (a) PU. (b) SA. (c) HU.

TABLE VII
EXPERIMENTAL RESULTS OF DIFFERENT α'S (%)

Dataset	Metric	0.93	0.95	0.97	0.99
PU	OA	96.18±2.15	**96.24±2.24**	95.51±2.35	95.43±2.68
	AA	94.98±0.74	**95.03±0.94**	94.70±0.44	94.45±0.69
	KC	95.52±2.72	**96.43±2.55**	95.42±2.99	95.22±3.40
SA	OA	95.37±0.63	**95.65±1.69**	94.38±1.20	94.49±0.89
	AA	98.22±0.19	**98.29±0.46**	97.96±0.33	97.99±0.31
	KC	94.86±0.70	**95.16±2.17**	93.76±1.33	93.88±0.98
HU	OA	95.38±0.39	**95.53±0.46**	95.33±0.35	95.37±0.42
	AA	94.91±0.32	**95.78±0.51**	95.80±0.38	95.91±0.35
	KC	95.00±0.42	**95.17±0.50**	94.95±0.38	95.00±0.45

According to the experimental results in Fig. 6, the optimal values of λ_D and λ_s for the PU, SA, and HU datasets are $\{\lambda_D = 0.1, \lambda_s = 0.5\}$, $\{\lambda_D = 0.1, \lambda_s = 10\}$, and $\{\lambda_D = 5, \lambda_s = 5\}$, respectively. In Fig. 6(a), when λ_s is greater than 1, the overall OA results obtained in the experiment are low on the PU dataset. In Fig. 6(b), when the value of λ_s is less than 5, the OA result gradually decreases with the increase of λ_D on the SA dataset. In Fig. 6(c), the OA results are relatively stable on the HU dataset.

The smoothing factor α in (35) affects the weight update process of N_s. In this article, experiments are set up to select the optimal α from [0.93, 0.95, 0.97, 0.99]. The experimental results on the three datasets are shown in Table VII.

As shown in Table VII, the optimal value of α for the PU, SA, and HU datasets is 0.95. Specifically, the best OA results of the three datasets are 96.24%, 95.65%, and 95.53%. When the α is either lower than 0.95 or higher than 0.95, the overall performance tends to decline. Meanwhile, the experimental results with α set to 0.93 demonstrate the stability of the EDSEL method.

In (33), to explore the appropriate form of temporal perturbation P_T, this article conducts experiments using various perturbation methods, including no perturbation, Gaussian perturbation, dropout perturbation, and *tanh* perturbation. Among them, the probability adopted for dropout perturbation is 0.5. The experimental results on the three datasets are shown in Table VIII.

As shown in Table VIII, when the *tanh* perturbation is adopted for P_T, the results on three datasets are the best.

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TABLE VIII
EXPERIMENTAL RESULTS OF DIFFERENT P_T'S (%)

Dataset	Metric	None	Gaussian	Dropout	Tanh
PU	OA	95.33±0.41	93.53±3.89	95.90±0.84	**96.24±2.24**
	AA	94.67±0.94	94.08±1.08	94.89±0.65	**95.03±0.94**
	KA	93.80±0.53	91.56±4.92	94.56±1.07	**96.43±2.55**
SA	OA	95.34±0.53	94.78±0.56	95.30±0.99	**95.65±1.69**
	AA	98.21±0.21	97.92±0.30	98.19±0.29	**98.29±0.46**
	KA	94.79±0.59	94.20±0.63	94.77±1.09	**95.16±2.17**
HU	OA	95.38±0.39	94.73±0.36	95.53±0.41	**95.53±0.46**
	AA	94.91±0.32	95.24±0.37	96.13±0.53	**95.78±0.51**
	KA	95.00±0.42	94.31±0.38	95.16±0.44	**95.17±0.50**

TABLE IX
RESULTS OF ABLATION EXPERIMENTS ON THREE DATASETS (%)

Dataset	Metric	EDSEL	w/o EASN	w/o TAL	w/o DGS
PU	OA	**96.24±2.24**	94.80±2.57	95.11±1.76	91.00±0.93
	AA	**95.03±0.94**	94.36±0.58	94.16±0.93	93.85±0.40
	KC	**96.43±2.55**	93.14±3.23	93.51±2.29	88.27±1.17
SA	OA	**95.65±1.69**	93.16±0.84	91.16±1.95	94.25±1.13
	AA	**98.29±0.46**	97.57±0.21	96.79±0.61	97.66±0.35
	KC	**95.16±2.17**	92.42±0.93	90.22±2.14	93.60±1.25
HU	OA	**95.53±0.46**	95.15±0.33	93.85±0.77	93.91±0.36
	AA	**95.78±0.51**	95.58±0.35	94.36±0.73	94.13±0.42
	KC	**95.17±0.50**	94.76±0.35	93.35±0.84	93.41±0.39

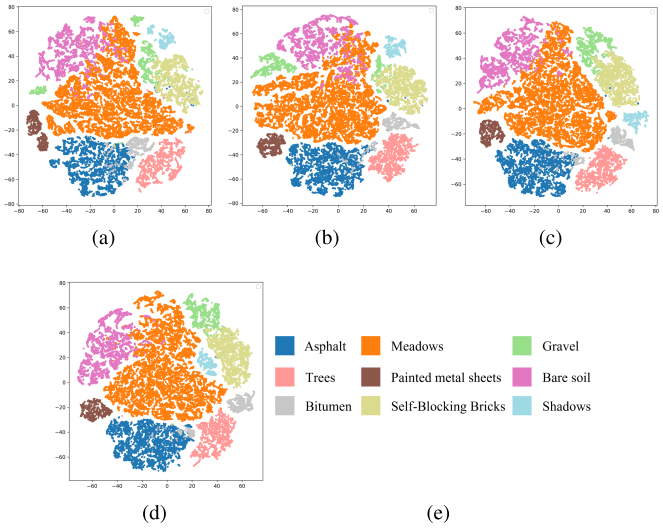


Fig. 7. t-SNE maps of the ablation study on the PU dataset. (a) Baseline model. (b) EASN. (c) EASN+TAL. (d) EASN+TAL+DGS. (e) Label color.

TABLE X
EXPERIMENTAL RESULTS OF CONSISTENCY LOSS ON THE PU DATASET (%)

Metric	w/o DGS	L_{PS}	L_{PS}+L_{PT}	L_{PS}+L_{PT} +LCP	L_{PS}+L_{PT} +LCP+LER
OA	91.00±0.93	93.03±1.49	93.57±2.67	94.39±1.72	96.24±2.24
AA	93.85±0.40	94.18±0.45	94.30±0.94	94.65±0.30	95.03±0.94
KC	88.27±1.17	90.92±1.89	91.49±3.43	92.64±2.19	96.43±2.55

F. Ablation Study

A series of experiments is conducted on PU, SA, and HU datasets to verify the effectiveness of the components in the EDSEL. Specifically, EASN, TAL, and DGS are individually removed from EDSEL each time, and the resulting OA, AA, and KC metrics are analyzed to assess the contribution of each component. Removing EASN means replacing the proposed spectral energy attention block U_S and spatial attention block U_C with standard convolution operations. When removing the TAL, the cross-entropy loss is considered. Removing the DGS means that the consistency regularization for unlabeled samples is removed. The results of these ablation experiments on the PU, SA, and HU datasets are presented in Table IX.

As shown in Table IX, removing EASN, TAL, or DGS results in a decrease in OA, AA, and KC across all three datasets, demonstrating the effectiveness of each component in EDSEL. On the PU dataset, removing DGS leads to 5.24% drop in OA, 1.18% in AA, and 8.16% in KC, indicating that DGS significantly enhances the model's classification capability by effectively leveraging unlabeled samples. This highlights the role of TAL in transferring knowledge from unlabeled learning to the student network, thereby improving its robustness.

Meanwhile, to verify the effectiveness of each component, the t-SNE algorithm is adopted in this article to visualize the features during the testing phase. Specifically, features are first visualized based on the basic convolutional network. Then, ESAN, TAL, and DSG are sequentially added, and the corresponding features are visualized. The relevant experimental results are shown in Fig. 7.

To verify the effectiveness of each component in the consistency loss of DGS, experiments are conducted on the PU dataset in this article. The first two terms in (38), L_{PS} and L_{PT}, are the consistency losses between the student and the teacher networks under static and temporal perturbations, respectively. The third term is the cross-perturbation consistency loss of the student network, abbreviated as L_{CP}. The fourth term is the entropy regularization loss, abbreviated as L_{ER}. The effectiveness of each term is evaluated based on the experimental results of OA, AA, and KC in Table X.

As shown in Table X, with the addition of different terms of the consistency loss, the OA, AA, and KC metrics on the PU dataset are all improved, which means that each term in (38) contributes to the model accuracy. Specifically, after adding static perturbation L_{PS}, OA increases from 91.00% to 93.03%, indicating that static perturbation can provide basic robustness constraints for the model. After adding temporal perturbation L_{PT} based on L_{PS}, OA rises from 93.03% to 93.57%, showing that the design of temporal perturbation is effective and can gradually strengthen the constraint without oscillation. Then, after introducing cross-perturbation consistency loss L_{CP}, OA is greatly improved to 94.39%, demonstrating that maintaining consistent predictions under different perturbations is crucial. Finally, after incorporating entropy regularization L_{ER}, OA

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TABLE XI
EXPERIMENTAL RESULTS OF ATTENTION MECHANISM ON THE PU DATASET (%)

Metric	Single attention	Attention reuse	Attention reuse and residual structure
OA	91.67±3.61	94.69±2.05	**96.24±2.24**
AA	94.08±0.82	94.40±0.80	**95.03±0.94**
KC	89.22±4.55	93.02±3.59	**96.43±2.55**

increases to 96.24% with the most significant improvement, which illustrates that entropy regularization can prevent the model from being overconfident and effectively address the core challenge of spectral heterogeneity in hyperspectral data.

To verify the effectiveness of the attention mechanism in the ESAN method, ablation experiments are carried out on the PU dataset. Experiments are conducted under three settings: using only single energy attention, using attention reuse, and using both attention reuse and residual structure. The experimental results are presented in Table XI.

As shown in Table XI, when using single energy attention, the OA is 91.67%, and the KC is 89.22%. After introducing the attention reuse mechanism, the OA is improved to 94.69% and the KC to 93.02%. This indicates that single energy attention can only capture shallow and local information, which is insufficient to fully model the complex band correlations and spatial dependencies. Through attention reuse, the model can repeatedly enhance the weights of highly discriminative neurons at different feature levels. After introducing the residual structure based on attention reuse, the OA is increased to 96.24% and the KC to 96.43%. This demonstrates that the residual structure can effectively alleviate the potential feature degradation problem and ensure that the deep attention module can still learn fine-grained information.

G. Stability Analysis

To demonstrate the effectiveness of EDSEL under the low-label regime, experiments are conducted on the PU dataset by selecting training samples with a ratio of 0.5%, and the results are shown in Table XII.

As shown in Table XII, with only 0.5% training data (fewer than 100 samples per class) on the PU dataset, EDSEL achieves the best performance in OA, AA, and KC. It outperforms the second-best method by 1.86%, 0.99%, and 1.92% on the three metrics, respectively. Under the low-label regime, RSEN and EDSEL show superior performance, as both semisupervised approaches can effectively utilize abundant unlabeled samples. Specifically, the DSG strategy in EDSEL applies effective perturbations in later training stages to learn discriminative spectral–spatial features of hyperspectral data.

The visualization results of seven state-of-the-art algorithms and EDSEL on the PU dataset under the low-label regime are shown in Fig. 8.

As shown in Fig. 8, under the low-label regime, most comparison algorithms suffer from obvious classification errors in their maps. As can be seen from Fig. 8(b) and (c), the classification results of ContextNet and DPyResNet exhibit

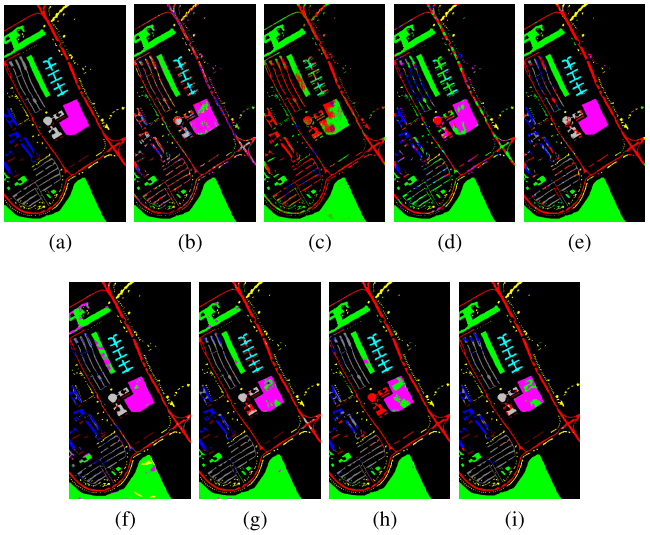


Fig. 8. Visualization maps of various algorithms on the PU dataset under the low-label regime. (a) Ground truth. (b) ContextNet. (c) DPyResNet. (d) SSRN. (e) A2S2KNet. (f) MLPA. (g) RSEN. (h) SSFEN. (i) EDSEL.

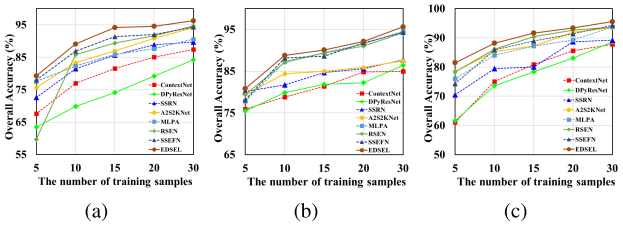


Fig. 9. OA results of three datasets with different training samples. (a) PU. (b) SA. (c) HU.

scattered noisy patches, which may be attributed to their poor ability to classify pixels with spectral heterogeneity in the PU dataset. Fig. 8(d) and (e) shows that SSRN and A2S2KNet tend to produce errors near the boundaries of different classes. From Fig. 8(f)–(h), MLPA, RSEN, and SSEFN also present misclassifications around class boundaries, mainly due to low classification confidence in complex mixed texture regions under the low-label condition. In contrast, the EDSEL algorithm ensures that the deep attention module learns fine-grained discriminative information via attention reuse and residual structures. It captures consistent spectral–spatial features with the DSG strategy, thus achieving superior performance in complex mixed regions.

Experiments with varying numbers of training samples are conducted on the three datasets to verify the stability of the proposed EDSEL algorithm. The number of training samples is set to [5, 10, 15, 20, 30] for the PU, SA, and HU datasets, and the performance of the proposed algorithm is compared with state-of-the-art methods to analyze stability. The experimental results under different training samples are shown in Fig. 9.

As shown in Fig. 9, EDSEL consistently achieves the best results across all three datasets, regardless of the number of training samples. Notably, when each class has only five training samples, EDSEL still outperforms other methods, demonstrating its ability to maintain high accuracy even with limited data. When the sample size is greater than 10, the

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TABLE XII
CLASSIFICATION RESULTS ON THE PU DATASET (%, 0.5% TRAINING SAMPLES)

Class	ContextNet	DPyResNet	SSRN	A2S2KNet	MLPA	RSEN	SSEFN	EDSEL
1	59.55±7.29	38.33±13.79	69.42±9.90	70.86±14.39	94.22±1.88	97.59±0.61	73.98±3.24	98.01±0.77
2	89.65±5.16	73.62±14.94	90.20±10.92	95.51±2.76	97.36±1.91	99.65±0.34	93.56±1.73	99.84±0.12
3	44.40±15.82	37.87±32.00	54.60±13.16	80.08±12.53	65.29±9.18	85.71±3.52	82.77±10.86	81.79±3.05
4	54.81±14.76	65.02±29.63	81.35±11.69	91.67±7.22	88.63±2.60	97.24±0.89	96.89±2.35	97.69±0.70
5	77.64±19.63	85.89±28.92	92.17±7.43	94.95±5.12	98.30±1.45	85.98±9.12	99.99±0.02	99.14±0.75
6	79.61±13.12	45.64±18.67	86.62±14.87	95.51±3.48	77.78±5.01	76.43±2.69	95.91±4.41	77.04±4.82
7	46.71±9.44	21.04±18.76	71.61±23.82	81.65±29.54	69.26±8.31	52.86±5.88	67.25±10.47	74.48±6.20
8	45.96±6.74	25.18±18.34	58.94±6.50	72.92±8.57	84.24±6.09	82.67±3.85	74.89±3.71	83.40±4.28
9	53.74±10.75	23.26±32.48	79.65±25.60	84.66±29.00	96.22±4.05	77.10±0.14	100.00±0.00	92.86±5.36
OA	74.41±5.30	56.95±7.32	77.81±8.41	85.82±6.86	89.37±1.63	92.03±0.55	87.64±1.66	93.46±0.74
AA	61.34±7.93	46.21±13.88	76.06±6.94	85.31±7.35	85.70±1.00	83.92±1.42	87.25±5.28	89.36±1.23
KC	65.13±7.55	37.05±15.45	70.11±10.84	80.89±9.29	87.10±1.33	89.30±0.74	83.28±2.31	91.22±1.01

TABLE XIII
RESULTS OF PARAMETERS, FLOPS, AND TIME COST ON THE SA DATASET

Metrics	ContextNet	DPyResNet	SSRN	A2S2KNet
Parameters (M)	0.56	21.81	0.37	0.08
FLOPs (G)	0.51	0.03	2.59	0.01
Training time (s)	9.06	2.75	58.89	25.94
Test time (s)	55.36	52.73	74.56	29.02
Metrics	MLPA	RSEN	SSEFN	EDSEL
Parameters (M)	0.05	1.34	16.50	1.35
FLOPs (G)	0.38	1.68	14.58	1.77
Training time (s)	4.50	0.06	1.81	2.60
Test time (s)	0.11	4.43	0.02	0.09

self-ensemble learning algorithms are always the top two in the OA results, which proves that consistent regulation on unlabeled samples can enhance the network's robustness.

H. Computational Complexity Analysis

To verify the computational complexity and experimental time of different algorithms, experiments are conducted on the SA dataset to evaluate the number of parameters, floating-point operations per second (FLOPs), training time, and test time of various algorithms. The experimental results are presented in Table XIII.

As shown in Table XIII, the DPyResNet has the largest number of parameters, while the SSRN has the smallest number of parameters; the SSEFN has the highest FLOPs, whereas the A2S2KNet has the lowest FLOPs. Neither the parameter counts nor the FLOPs of EDSEL are excessively large, which is because the EASN module does not introduce additional network parameters during training. In terms of training and testing time, SSRN and A2S2KNet require longer training time, while MLPA, SSEFN, and EDSEL have shorter testing time.

IV. CONCLUSION

This article proposes the EDSEL algorithm to address the challenges of limited labeled samples and single perturbation during training in HSIC. First, an energy attention-based self-ensembling network is introduced, leveraging neuron activation mechanisms and residual connections to optimize the spectral energy attention block and the spatial attention block, thereby extracting more discriminative features. During self-ensembling learning, the consistency between student and teacher networks in feature learning and prediction results enhances model stability. In the training process with labeled samples, the teacher network guides feature learning by aligning the knowledge between the student and teacher networks. A DGS for unlabeled samples is designed to facilitate consistency regularization by applying static and temporal noise for data augmentation, strengthening perturbations in the later training stages, and improving network robustness and generalization. The proposed method is compared with six state-of-the-art HSIC algorithms on the PU, SA, and HU datasets. The results show that EDSEL achieves superior OA compared to other methods, reaching 96.24%, 95.65%, and 95.53% on the three datasets. Moreover, EDSEL consistently outperforms other methods in experiments with varying numbers of training samples, demonstrating its stability. However, there are still some areas for improvement. For instance, the teacher network updates its weights using the EMA method, which lacks adaptivity. The adaptive weights between the teacher and student networks may accelerate the convergence. In addition, the current consistency learning process applies the same operations to all samples, and introducing differentiated learning strategies based on sample confidence levels is a potential direction for future research.

DECLARATION OF COMPETING INTEREST

The authors declared that they have no conflicts of interest in this work.

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